CO-1

ASSESSMENT TOOL-1

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1. The Company wants to automate its payroll process Write a C program to calculate the total Salary by adding the basic Salary and bonus amount of an employee.

Pseudo Code :

Start

Read Basic Salary , Bonus

Total Salary = Basic Salary + Bonus

Print Total Salary

Stop

Input :

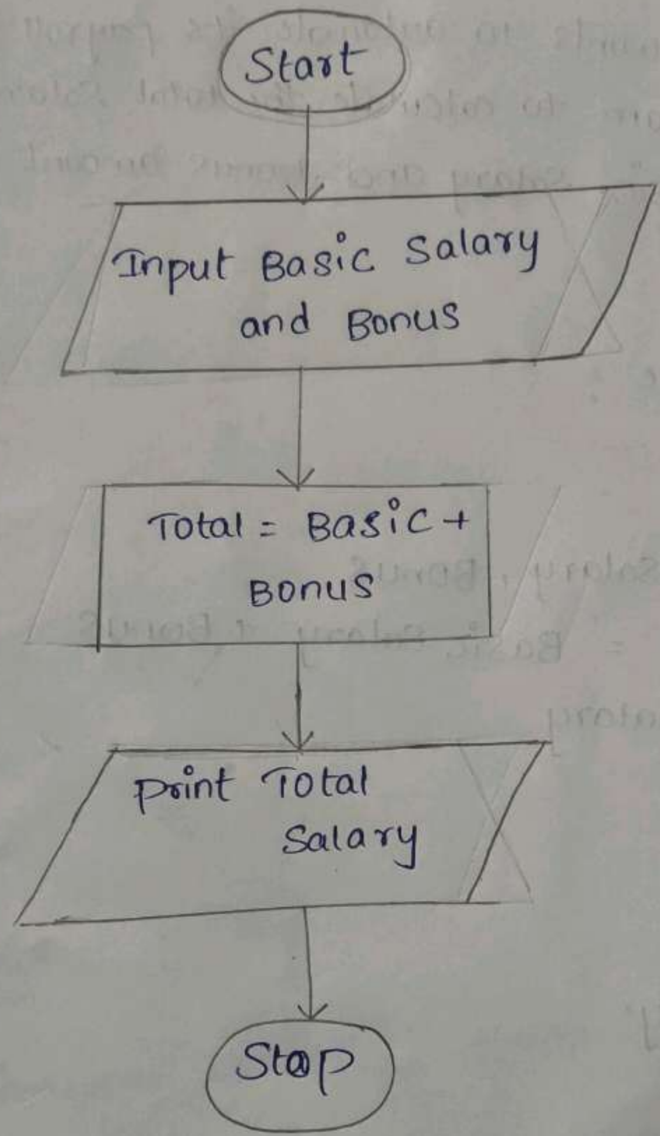
Basic Salary

Bonus

Output :

Total Salary

Flow chart :



2. During a database update, two employee IDs need to be exchanged. Write a C program to Swap two numbers using a temporay variable.

Pseudo Code :

Start

Read A, B

Temp = A

A = B

B = Temp

print A, B

STOP

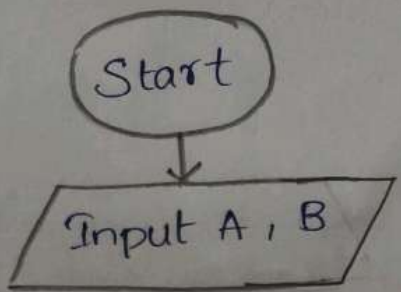
Input :

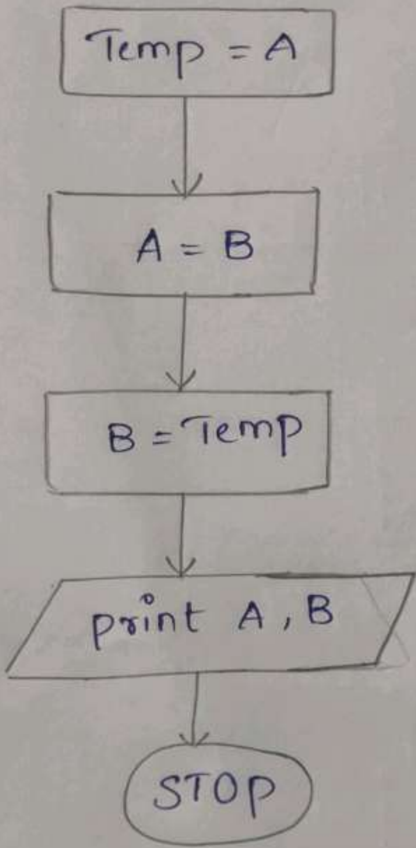
- A
- B

Output :

- Swapped values OF A and B

Flowchart :





3. In an embedded System with limited memory two values must be exchanged without using any extra Storage. Write a c program to Swap two numbers without using two numbers.

Pseudo code :

START
Read A, B
A = A + B
B = A - B

A = A - B

print A, B

STOP

Input :

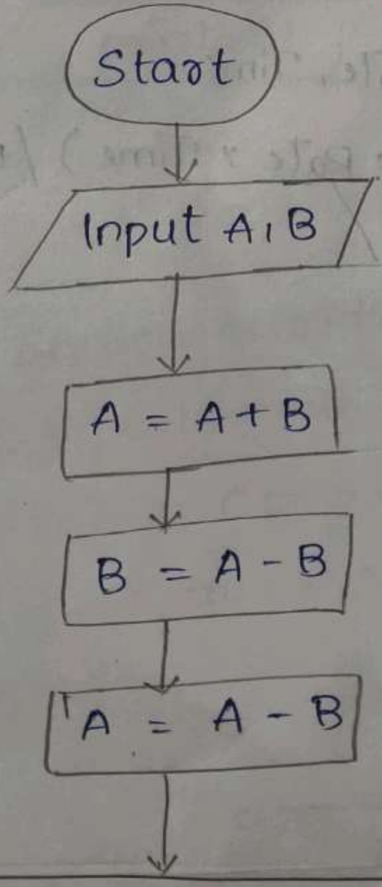
A

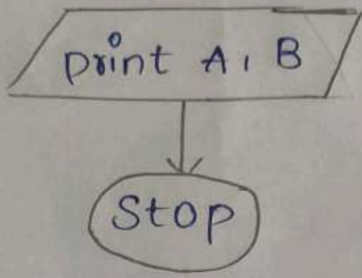
B

Output :

Swapped values of A and B

Flowchart :





4. A bank wants to provide customers with an estimate of the interest earned on their deposit. Write a C program to calculate simple interest using the formula.

pseudo code :

Start
Read principal, Rate, Time
SI = (principal × Rate × Time) / 100
print SI
STOP

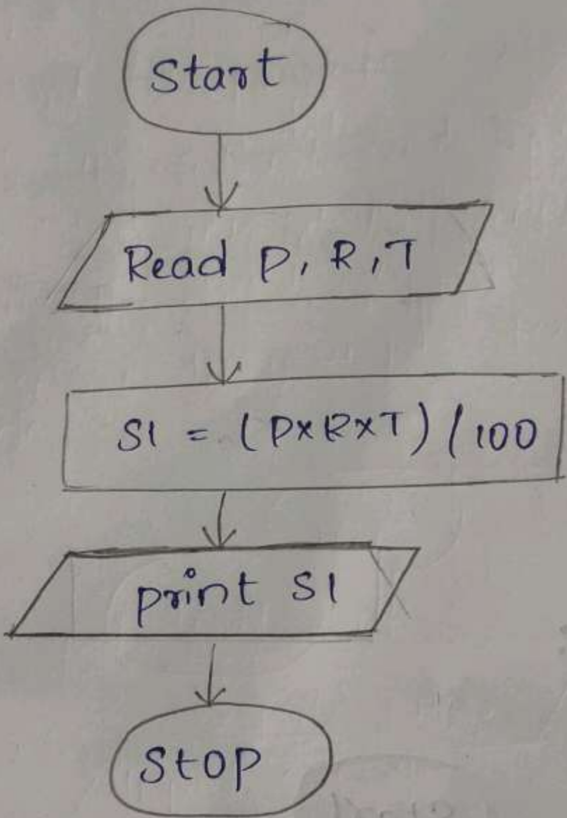
Input :

principal (P)
Rate (R)
Time (T)

Output :

Simple Interest

Flowchart :



5. A Weather Station records temperature in Celsius, but reports need to be generated in Farenheit. Write a C program to convert tempera-ture From Celsius to Fahrenheit

Pseudo code :

Start

Read Celsius

Fahrenheit = (celsius x 9/5) + 32

print Fahrenheit

stop

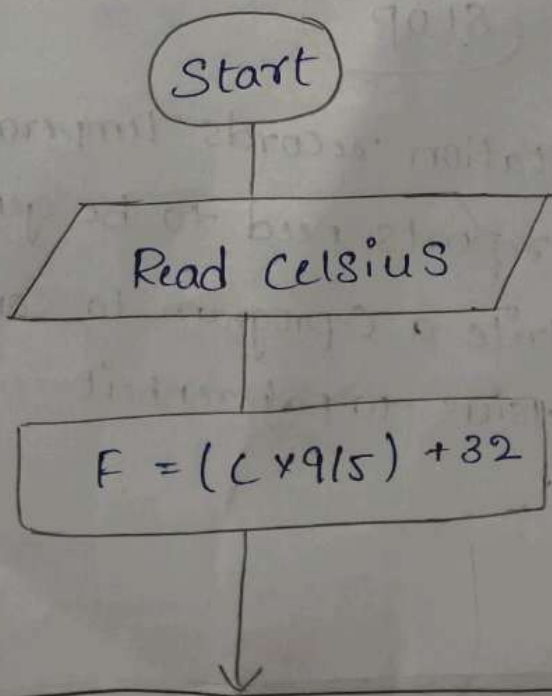
Input :

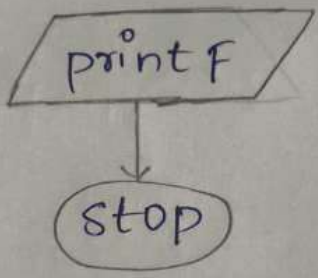
Celsius

Output :

Fahrenheit

Flow Chart :





6. A mathematics learning application need to display both the quotient and remainder when two numbers are divided. Write a C programming to find the quotient and remainder of two integers.

Pseudo Code :

Start
Read A, B
Quotient = A/B
Remainder = A%.B
Print quotient
Print Reminder
stop

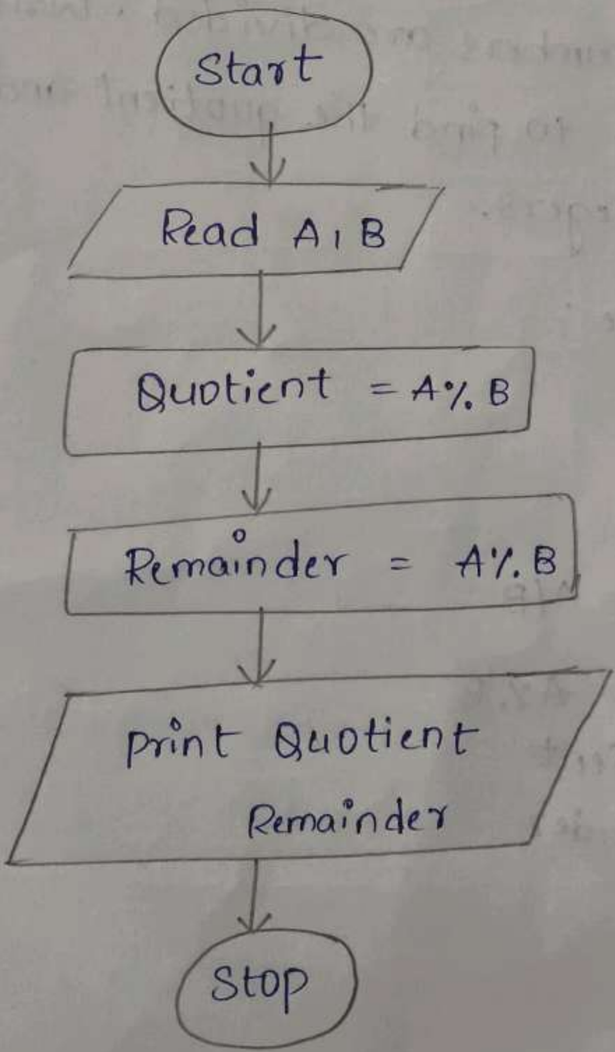
Input :

- Divident (A)
- Divisor (B)

Output :

- Quotient
- Remainder

Flowchart :-



7. An event management system allows entry based on even-numbered ticket IDS. Write a C program to determine whether a given ticket number is even or odd.

Pseudo code

Start

Read N

If N % 2 == 0

print "Even"

Else

print "odd"

stop

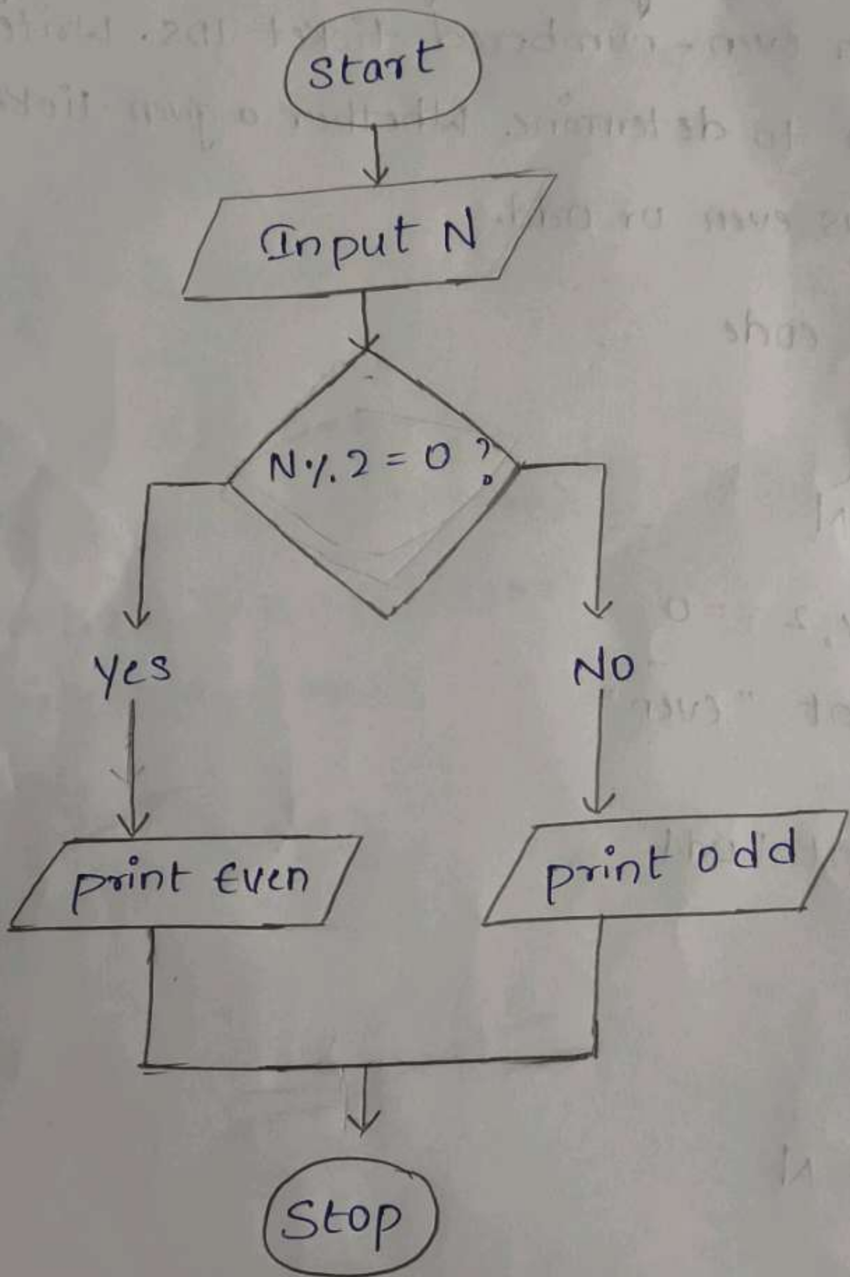
Input:

- integer N

Output:

- Even or odd

Flow Chart :-



8 An engineer needs to solve quadratic equations OF the From. Write a C program to calculate and display the roots OF the quadratic equations

pseudo code :

Start

Read ~~N~~ a, b, c

If N%2==0

print

D = b*b - 4*a*c

Root1 = (-b + √D) / (2*a)

Root2 = (-b - √D) / (2*a)

print Root1, Root2

STOP

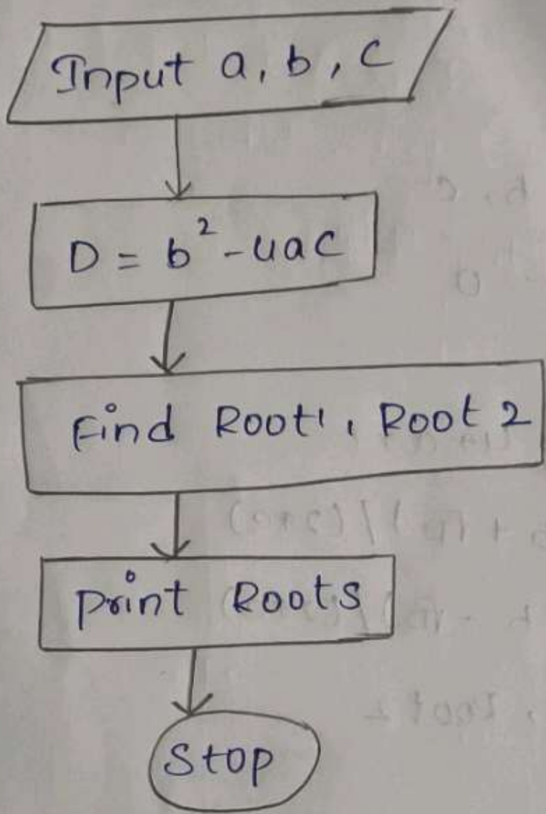
Input :-

- a, b, c

Output :-

- Root1 and Root2

Flow chart :-



9. A scientific calculator application requires the squrae and cube of a given number. Write a C program to calculate the Square and Cube of a number entered by the user.

Pseudo code :

Start

Read N

Square = N * N

Cube = N * N * N

print Square, cube

Stop

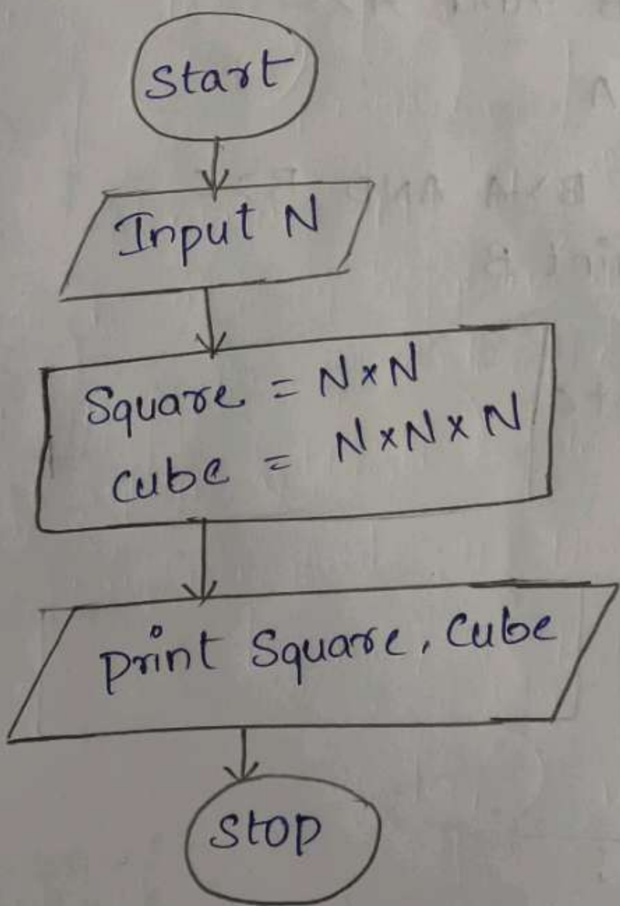
input :

Number N

Output

- Square and cube

Flow chart :



10. A Company wants to identify the top-performing salesperon among three employees based on their Sales Figures write a c program to Find the largest among the three numbers.

Pseudo code :-

Start

Read A, B, C

If A > B And A > C

print A

ElSE If B > A AND B > C

print B

ElSE

print C

Stop

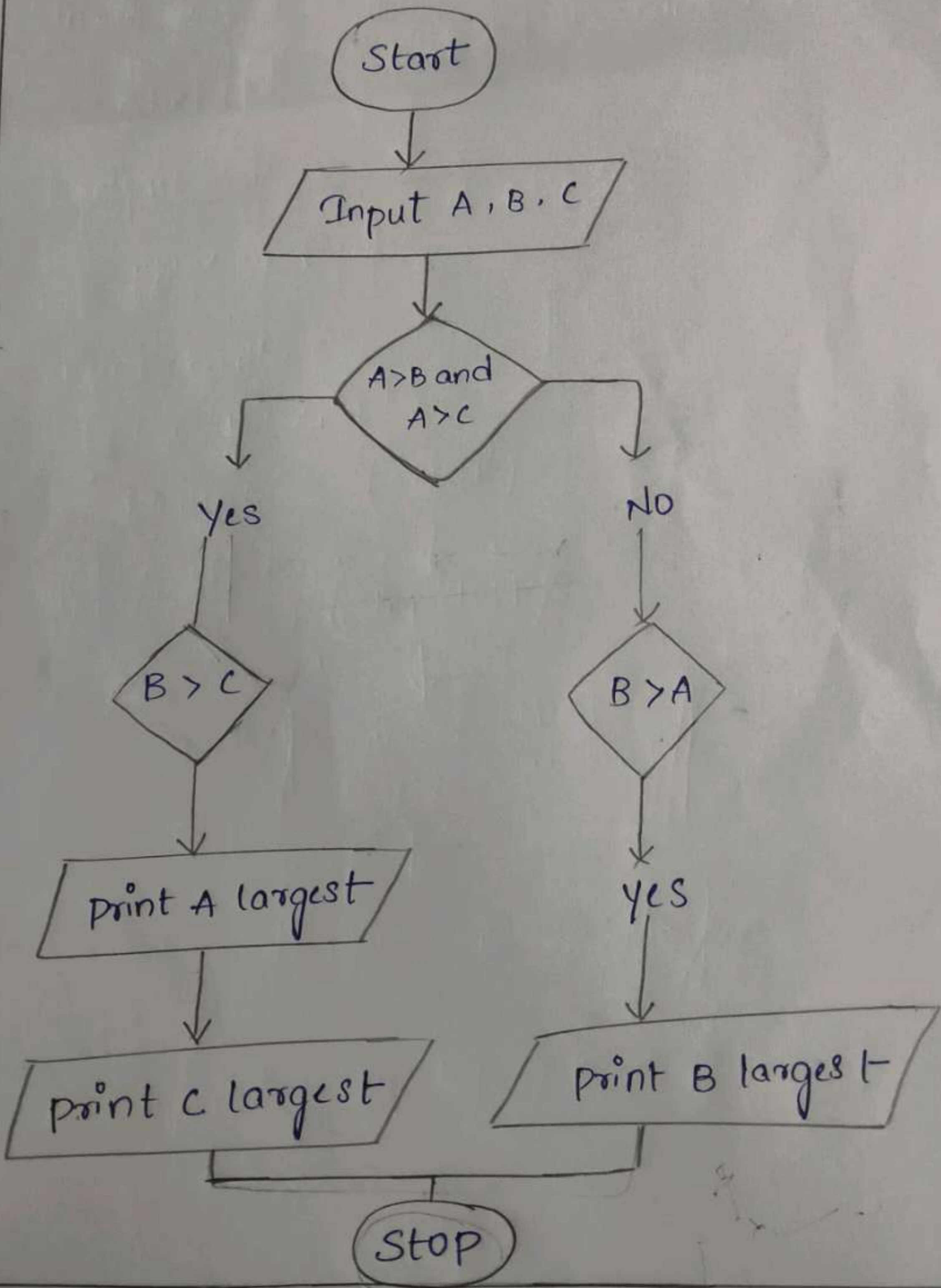
input :

- A, B, C

Output :-

- largest number

Flow Chart :



Start
Input A, B, C
A>B and A>C
Yes
NO
B > C
B >A
Print A largest
Yes
Print C largest
Print B largest
Stop